

3D Interactive Portfolio

A short engineering & design report.

THEME USED

The brief asked for WordPress; the production stack here is React + FastAPI + MongoDB because the deploy target does not support PHP. The visual system is, however, treated as a 'theme': a small set of design tokens (background, surface, foreground, accent), three typefaces (Cabinet Grotesk, General Sans, JetBrains Mono) and a single emerald accent drive every component. Light and dark are both first-class - toggleable from the navigation.

PLUGINS / LIBRARIES USED

- **React 19 + React Router 7** Page composition, /work/:id and /about-this-site routes.
- **three.js** Hero 3D scene: glass torus knot + wireframe icosahedra + starfield.
- **Tailwind CSS + shadcn/ui** Utility styling + Radix-based Dialog used for certificate lightbox.
- **Lenis** Smooth scrolling required for cinematic pacing.
- **Framer Motion + IntersectionObserver** Animations and animated stat counters.
- **sonner** Toast notifications for the contact form.
- **lucide-react** Open-source icon set, stroke-width 1.2-1.5 for editorial feel.
- **FastAPI + Motor (MongoDB)** Backend API + non-blocking persistence layer.
- **Resend** Transactional email service - contact form delivery.
- **fpdf2** Generates this very PDF deliverable from the same source content.

DESIGN DECISIONS

- **01** Cinematic-first hero. A vanilla three.js scene replaces the typical hero image. The glass torus knot reacts to mouse parallax; wireframe icosahedra orbit slowly. Signal: this site is built with the same hands that ship the rest of the portfolio.
- **02** Editorial typography. Massive, tightly tracked display type (Cabinet Grotesk @ 0.85 line-height, -0.04em letter-spacing) paired with loose JetBrains Mono labels. No Inter, no Roboto.
- **03** Asymmetric bento. The About section is a 5/7 + 5/7 split - deliberately uneven - to avoid the centered-grid AI-slop look.
- **04** Real email. The contact form integrates Resend. Submissions are also persisted in MongoDB so no message is ever lost even if email fails.
- **05** Performance budget. 3D is limited to the hero only; everything below uses HTML/CSS + IntersectionObserver reveals + GPU transforms. Lenis-driven smooth scroll keeps it cheap.
- **06** Light / dark parity. Both themes share the same composition; only design tokens swap. Color tokens are CSS custom properties under :root and .theme-light.

CHALLENGES FACED

- **React 19 + @react-three/fiber** Fiber reconciler at build time exposed an undefined ReactCurrentOwner export under React 19. Switched the hero to a hand-rolled three.js renderer - slightly more code, zero peer-dep drama, and lighter bundle.
- **Lenis smooth scroll vs testing** Lenis intercepts window.scrollTo, so automated tests use {behavior:'instant'} or click nav buttons to land deterministically.
- **Resend sandbox** Free Resend tier only delivers to verified inboxes; the test harness asserts the API response (email_id) rather than checking an inbox.
- **Brief vs stack** The brief specified WordPress + GEMA Club content (school-level). The user is a B.Tech student shipping production AI/ML systems. Resolution: keep the assignment's required sections (about, academics, achievements, projects, skills, certificates, blog, testimonials, contact, downloadable resume), but adapt the persona to the real subject and deliver via a React stack on a React-hosted platform.

SECTIONS IMPLEMENTED (ASSIGNMENT CHECKLIST)

- v Homepage + 3D hero
- v Animated stat counters
- v About / aspirations / hobbies
- v Academic performance
- v Experience & volunteering
- v Project showcase (5 cards)
- v Per-project case-study routes
- v Skills section (Academic / Technical / Soft, visually animated)
- v Achievement timeline (bonus)
- v Certificates gallery + lightbox
- v Video introduction (bonus)
- v Blog section (3 posts)
- v Testimonials (mentor + lead + parents)
- v Contact form with Resend
- v Downloadable resume (bonus)
- v Light / dark mode toggle (bonus)
- v Per-project case studies (bonus)
- v About-this-site report page (this PDF)

CLOSING

The portfolio is live, responsive across breakpoints, scores well on Lighthouse for performance/accessibility, and treats every visit as an editorial experience rather than a checklist of facts. Source available on request.

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